

Roadmap

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Stage 1 — Day 1: Introduction to Python Development

- What is CLI (Command Line Interface)
- Difference between Python 2 and Python 3
- Introduction to Django
- PEP 8 Python style guide
- GIGO (Garbage In, Garbage Out)
- Introduction to Anaconda

2A

Stage 2 — Day 2: CLI Fundamentals

- Detailed introduction to CLI
- What is CLI and why developers use it
- Common CLI operations
- Useful CLI tips
- Relative paths vs absolute paths
- Running Python from the terminal

2B

Stage 3 — Day 2: Git Fundamentals

- What is Git and why it is used
- Basic Git operations
- Initialize a Git repository
- Check file status
- Add and commit changes
- View commit history
- Push and pull project changes

3A

Stage 4 — Day 3: PowerShell Automation

- Introduction to automation
- PowerShell scripts with the .ps1 extension
- Run .ps1 files from the terminal
- Use Task Scheduler
- Create automatic backups
- Use Write-Host to display messages
- Create folders and files automatically

3B

Stage 5 — Day 3: Python Project Setup and Git Branching

- Automate Python project setup using a .ps1 script
- Create and activate a Python virtual environment
- Install required packages
- Save dependencies using pip freeze
- Initialize Git automatically
- Basic Git branching
- Merge branches into the main branch

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Stage 6 — GitHub and Collaboration

- Introduction to GitHub
- Connect a local Git repository to GitHub
- Configure Git username and email
- Create and manage remote repositories
- Push local commits to GitHub
- Pull changes from GitHub
- Create and switch between branches
- Merge branches
- Fork an existing repository
- Contribute to projects using the Fork → Clone → Edit → Push → Pull Request workflow